Exploratory Testing Session Notes: Projects API

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Charter

- Identify capabilities and areas of potential instability of the rest api todo list manager.
- Identify documented and undocumented rest api todo list manager capabilities.
- For each capability, create a script or small program to demonstrate the capability.
- Exercise each capability identified with data typical to the intended use of the application.
- Scope of this session: the projects endpoints. Timebox: 45 minutes.
- Environment: runTodoManagerRestAPI-1.5.5.jar on localhost:4567, driven with curl from bash. Server on Java 19.
- Files: Exploratory testing scripts/Projects/projects_session.sh and projects_session_output.txt.

Session notes

Endpoint	Method	Documented?	Expected behavior	Match	Observed behavior	Side notes	Files
/projects	GET	DOCUMENTED	200, every project	YES	200, all projects under a "projects" key, JSON by default	List order changes between calls	projects_session_output.txt: /projects. ProjectsTest
/projects	GET	DOCUMENTED	XML when Accept is application/xml	YES	200, <projects> document		projects_session_output.txt: /projects. ProjectsTest
/projects	GET	DOCUMENTED	Filter by field=value query params	YES	?active=true and ? title=..&completed=.. return only matching projects	Unknown param is ignored, returns everything	projects_session_output.txt: /projects. ProjectsTest
/projects	HEAD	DOCUMENTED	Headers for all projects	YES	200, Content-Type application/json, no body		projects_session_output.txt: /projects. ProjectsTest
/projects	POST	DOCUMENTED	Create project from body fields, no id	YES	201, new id, fields stored. JSON and XML bodies both work	No field is mandatory: {} creates an empty project	projects_session_output.txt: /projects. ProjectsTest
/projects	POST	DOCUMENTED	Documented example JSON input is accepted	NO	400 "completed should be BOOLEAN, active should be BOOLEAN"	Docs show booleans as strings. Real JSON booleans work. Bug PROJ-004	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest

Endpoint	Method	Documented?	Expected behavior	Match	Observed behavior	Side notes	Files
/projects	POST	DOCUMENTED	Documented example XML input is accepted	NO	400 "Not allowed to create with id"	Example contains <id>null</id>. Bug PROJ-004	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest
/projects	POST	UNDOCUMENTED	Body with a tasks array	N/A	201, project created already linked to the listed todos	Decided to cover in unit tests	projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects	OPTIONS	UNDOCUMENTED	Allowed methods	N/A	200, Allow: OPTIONS, GET, HEAD, POST		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects	PUT	UNDOCUMENTED	method not allowed	YES	405, empty body		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects	DELETE	UNDOCUMENTED	method not allowed	YES	405, empty body		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects/:id	GET	DOCUMENTED	200, the project with that id	YES	200, one project wrapped in a "projects" array. XML also works	Missing or non numeric id gives 404 with errorMessages	projects_session_output.txt: /projects/:id. ProjectByIdTest
/projects/:id	HEAD	DOCUMENTED	Headers for the project	YES	200, no body. 404 for a missing id		projects_session_output.txt: /projects/:id. ProjectByIdTest
/projects/:id	POST	DOCUMENTED	Amend the fields in the body	YES	200, supplied fields change, other fields and links kept	id in the body gives 400	projects_session_output.txt: /projects/:id. ProjectByIdTest
/projects/:id	PUT	DOCUMENTED	Amend the fields in the body	NO	200, but omitted fields reset to defaults and all task and category links are removed	Data loss. Bug PROJ-001	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest
/projects/:id	DELETE	DOCUMENTED	Delete the project	YES	200, later GET is 404. Its todos survive and lose the tasksof link	Second delete gives 404	projects_session_output.txt: /projects/:id DELETE. ProjectByIdTest, ProjectsInvalidTest
/projects/:id	OPTIONS	UNDOCUMENTED	Allowed methods	N/A	200, Allow: OPTIONS, GET, HEAD, POST, PUT, DELETE		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects/:id	PATCH	UNDOCUMENTED	method not allowed	YES	405, empty body		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest

Endpoint	Method	Documented?	Expected behavior	Match	Observed behavior	Side notes	Files
/projects/:id/tasks	GET	DOCUMENTED	Todos linked to the project	YES	200, "todos" array, empty for a new project. XML also works		projects_session_output.txt: /projects/:id/tasks. ProjectTasksTest
/projects/:id/tasks	GET	DOCUMENTED	404 for a project that does not exist	NO	200 with the tasks of every other project, with duplicates	Wrong data returned. Bug PROJ-002	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest
/projects/:id/tasks	HEAD	DOCUMENTED	Headers for the tasks	YES	200, no body	Also 200 for a missing project (PROJ-002)	projects_session_output.txt: /projects/:id/tasks. ProjectTasksTest
/projects/:id/tasks	POST	DOCUMENTED	Link the todo whose id is in the body	YES	201, empty body. Todo shows the project under tasksof	Repeat gives 201, no duplicate. Missing todo or project gives 404. Numeric id {"id":1} gives 404	projects_session_output.txt: /projects/:id/tasks. ProjectTasksTest, ProjectsInvalidTest
/projects/:id/tasks	POST	DOCUMENTED	Link an existing todo with an XML body	NO	404 "Could not find thing matching value for id" for <todo><id>..</id></todo>	Same on /categories. JSON works. Bug PROJ-005	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest
/projects/:id/tasks	POST	UNDOCUMENTED	Body with todo fields and no id	N/A	201, creates a new todo and links it to the project	Decided to cover in unit tests	projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects/:id/ tasks/:id	DELETE	DOCUMENTED	Remove the link only	YES	200, link gone on both sides, todo still exists	Second delete gives 404	projects_session_output.txt: /projects/:id/tasks/:id. ProjectTasksTest, ProjectsInvalidTest
/projects/:id/ tasks/:id	DELETE	DOCUMENTED	404 for a project that does not exist	NO	400 with a null pointer error ("parent" is null) when the todo belongs to another project	Internal error leaks. Text varies with the Java version. Bug PROJ-003	projects_session_output.txt: Behavior that differs. ProjectsDocumentedBehaviorTest, ProjectsActualBehaviorTest
/projects/:id/ categories	GET	DOCUMENTED	Categories linked to the project	YES	200, "categories" array. XML also works	Missing project returns other projects' categories (PROJ-002)	projects_session_output.txt: /projects/:id/categories. ProjectCategoriesTest
/projects/:id/ categories	HEAD	DOCUMENTED	Headers for the categories	YES	200, no body		projects_session_output.txt: /projects/:id/categories. ProjectCategoriesTest
/projects/:id/ categories	POST	DOCUMENTED	Link the category whose id is in the body	YES	201, project lists the category. Category itself is unchanged	One way only: /categories/:id/projects stays empty	projects_session_output.txt: /projects/:id/categories. ProjectCategoriesTest

Endpoint	Method	Documented?	Expected behavior	Match	Observed behavior	Side notes	Files
/projects/:id/ categories	POST	UNDOCUMENTED	Body with category fields and no id	N/A	201, creates a new category and links it		projects_session_output.txt: Undocumented. ProjectsUndocumentedTest
/projects/:id/ categories/:id	DELETE	DOCUMENTED	Remove the link only	YES	200, link gone, category still exists	Second delete gives 404	projects_session_output.txt: /projects/:id/categories/:id. ProjectCategoriesTest, ProjectsInvalidTest
/projects	POST	DOCUMENTED	Malformed JSON is rejected	YES	400, errorMessages names a java.io.EOFException, nothing created	Content-Type of the error is text/html	projects_session_output.txt: Invalid operations. ProjectsInvalidTest
/projects	POST	DOCUMENTED	Malformed XML is rejected	YES	400 "Mismatched title and project", nothing created	Content-Type of the error is text/html	projects_session_output.txt: Invalid operations. ProjectsInvalidTest
/projects	POST	DOCUMENTED	Invalid bodies are rejected	YES	400 for an id, an unknown field or a non boolean flag. 415 for form data. 406 for Accept text/plain	Some rejected creates still use up an id	projects_session_output.txt: Invalid operations. ProjectsInvalidTest

Summary of session findings

- Projects support create, read, amend and delete in both JSON and XML.
- No project field is mandatory. Booleans must be real JSON booleans on input and come back as strings.
- GET /projects can be filtered by any field with query parameters.
- tasks links a project to todos in both directions (the todo shows tasksof).
- categories links a project to categories in one direction only.
- Linking the same todo or category twice does not create a duplicate.
- Deleting a project or a link never deletes the todo or category on the other end.
- Undocumented: OPTIONS, 405 on unsupported methods, creating a todo or category through a relationship POST, tasks inside a project POST body.
- Missing projects, todos and categories give 404 with an errorMessages body in most cases.

Summary of concerns

- PUT /projects/:id wipes omitted fields and every task and category link (PROJ-001).
- GET and HEAD on /projects/:id/tasks and /categories answer 200 with other projects' data when the project does not exist (PROJ-002).
- DELETE /projects/:id/tasks/:id on a missing project returns 400 with a raw null pointer error (PROJ-003).
- The example JSON and XML inputs in the documentation are rejected with 400 (PROJ-004).
- An existing todo or category cannot be linked with an XML body, only with JSON (PROJ-005).

- Malformed payload errors are sent as text/html although the body is JSON.
- Project to category links are invisible from the category side.
- An id sent as a JSON number in a relationship body is treated as not found.

Test ideas

- Check that every amend and link leaves all other todos, projects and categories untouched.
- Repeat the missing parent checks on /todos/:id/tasksof and /categories/:id/projects.
- Compare PUT and POST amend on todos to see if links are lost there too.
- Combine several query filters and filter on empty strings.
- Send very long titles and non ASCII text.
- Create and delete many projects to see how ids are allocated after rejected requests.
- Try XML relationship bodies on /todos/:id/tasksof and /todos/:id/categories.